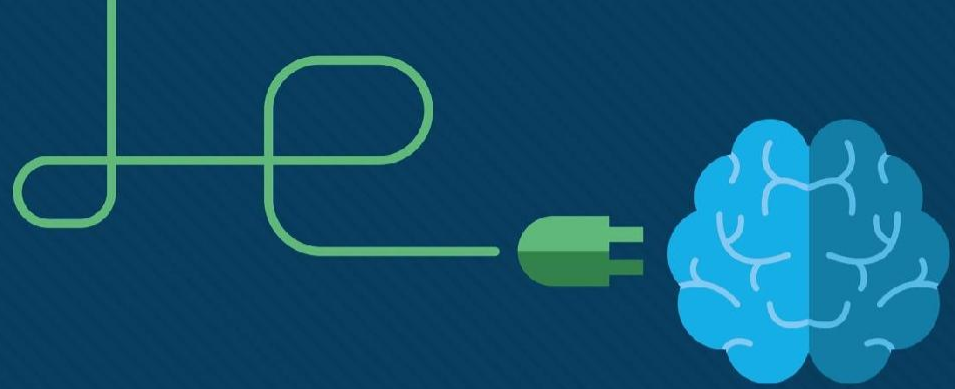




Chapter 5: Ethernet

Introduction to Networks v7.02
(ITN)



Chapter 5 - Sections & Objectives

■ 5.1 Ethernet Protocol

- Explain the operation of Ethernet.
- Explain how the Ethernet sublayers are related to the frame fields.
- Describe the Ethernet MAC address

■ 5.2 LAN Switches

- Explain how a switch operates.
- Explain how a switch builds its MAC address table and forwards frames.
- Describe switch forwarding methods and port settings available on Layer 2 switch ports.

■ 5.3 Address Resolution Protocol

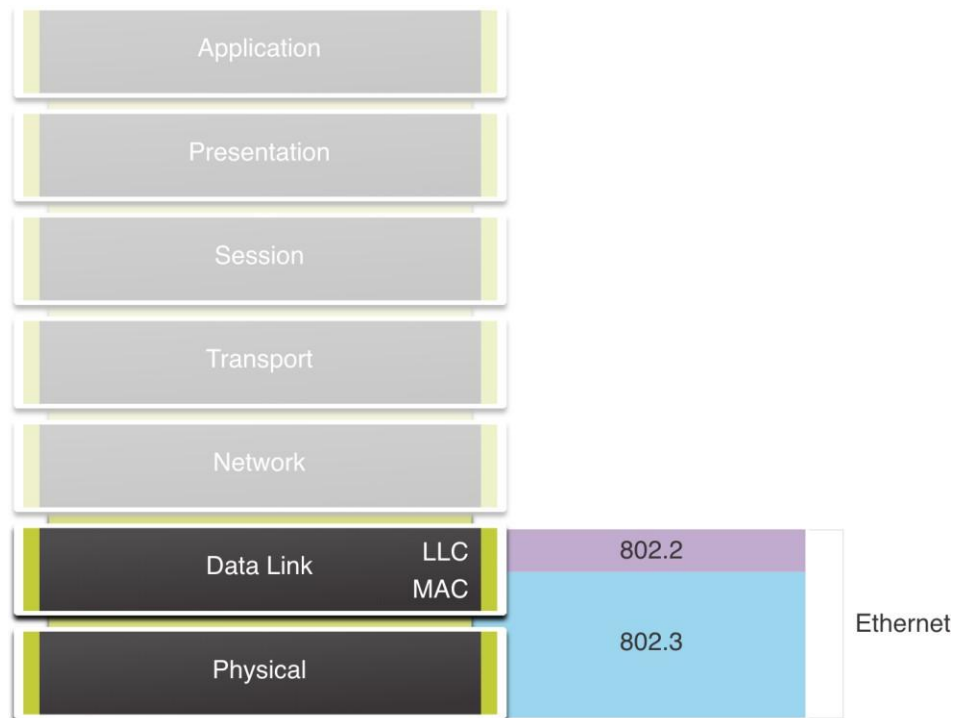
- Explain how the address resolution protocol enables communication on a network.
- Compare the roles of the MAC address and the IP address.
- Describe the purpose of ARP.
- Explain how ARP requests impact network and host performance.

5.1 Ethernet Protocol

Ethernet Frames

Ethernet Encapsulation

- Ethernet operates in the data link layer and the physical layer.
- It is a family of networking technologies defined in the IEEE 802.2 and 802.3 standards.

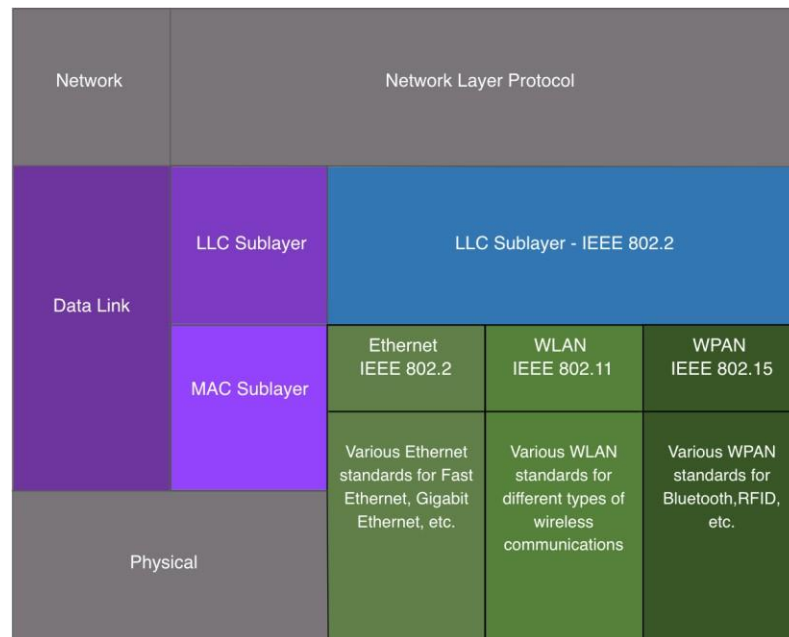


Ethernet Frames

Data Link Sublayers

The 802 LAN/MAN standards, including Ethernet, use two separate sublayers of the data link layer to operate:

- **LLC Sublayer:** (IEEE 802.2) Places information in the frame to identify which network layer protocol is used for the frame.
- **MAC Sublayer:** (IEEE 802.3, 802.11, or 802.15) Responsible for data encapsulation and media access control, and provides data link layer addressing.



Ethernet Frames

MAC Sublayer

The MAC sublayer is responsible for **data encapsulation** and **accessing the media**.

Data Encapsulation

IEEE 802.3 data encapsulation includes the following:

1. **Ethernet frame** - This is the internal structure of the Ethernet frame.
2. **Ethernet Addressing** - The Ethernet frame includes both a source and destination MAC address to deliver the Ethernet frame from Ethernet NIC to Ethernet NIC on the same LAN.
3. **Ethernet Error detection** - The Ethernet frame includes a frame check sequence (FCS) trailer used for error detection.

Media access control

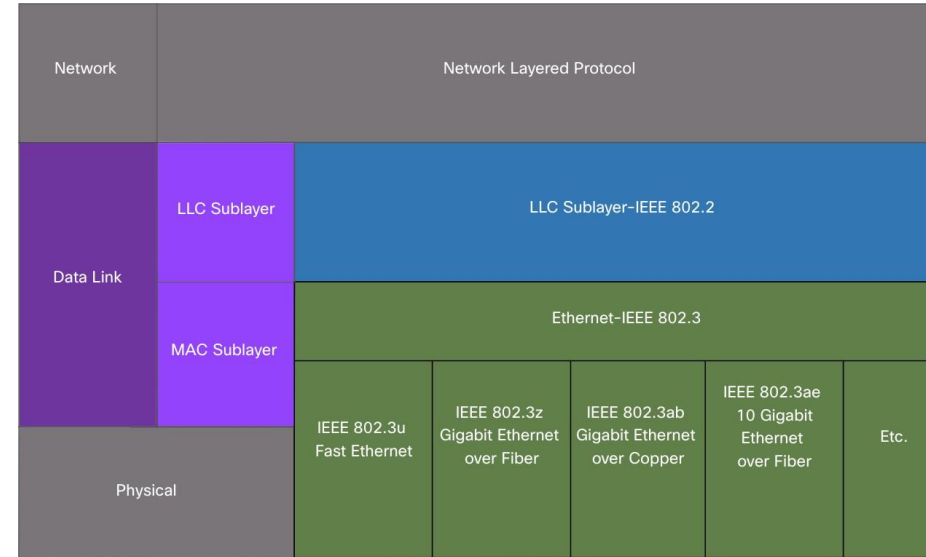
- (i) **responsible for the placement of frames on the media and the removal of frames from the media**, and (ii) **media recovery**. This sublayer communicates directly with the physical layer.

Ethernet Frames

MAC Sublayer

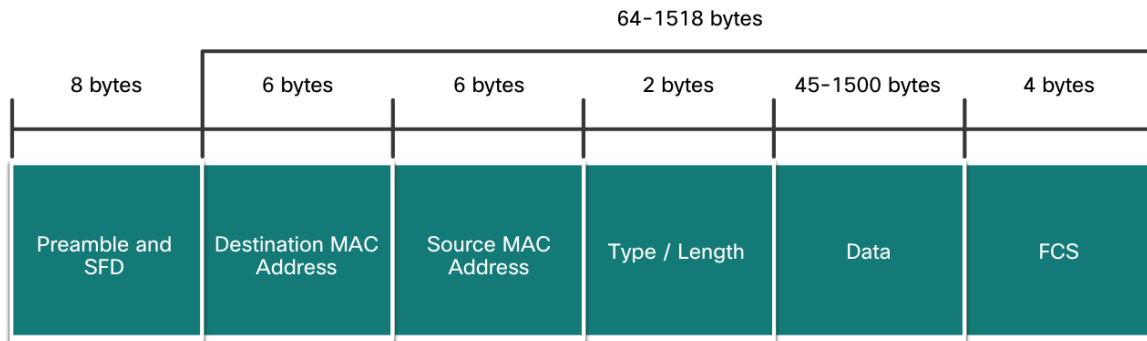
Media Access

- The IEEE 802.3 MAC sublayer includes the specifications for different Ethernet communications standards over various types of media including copper and fiber.
- Legacy Ethernet using a bus topology or hubs, is a shared, half-duplex medium. Ethernet over a half-duplex medium uses a contention-based access method, carrier sense multiple access/collision detection (CSMA/CD).
- Ethernet LANs of today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.



Ethernet Frame Fields

- The **minimum Ethernet frame size is 64 bytes and the maximum is 1518 bytes**. The preamble field is not included when describing the size of the frame.
- Any frame less than 64 bytes in length is considered a “**collision fragment**” or “**runt frame**” and is automatically discarded. Frames with more than 1500 bytes of data are considered “**jumbo**” or “**baby giant frames**”.
- If the size of a transmitted frame is less than the minimum, or greater than the maximum, the receiving device **drops the frame**. Dropped frames are likely to be the result of collisions or other unwanted signals. They are considered invalid. Jumbo frames are usually supported by most Fast Ethernet and Gigabit Ethernet switches and NICs.



Introduction to the Ethernet Frame

IEEE 802.3

7	1	6	6	2	46 to 1500	4
Preamble	Start of Frame Delimiter	Destination Address	Source Address	Length	802.2 Header and Data	Frame Check Sequence

Preamble
Notify destination to get ready for the new frame.

Start Frame Delimiter Fields
Used for synchronization between the sending and receiving devices

Length/Type Field
Defines the exact length of the frame's data field/ describes which protocol is implemented

802.2 Header & Data
Increase this frame field to at least 64 bytes

Introduction to the Ethernet Frame

IEEE 802.3

7	1	6	6	2	46 to 1500	4
Preamble	Start of Frame Delimiter	Destination Address	Source Address	Length	802.2 Header and Data	Frame Check Sequence

Frame Check Sequence Field

Used to detect errors in a frame with cyclic redundancy check (4 bytes), if calculations match at source and receiver, no error occurred.

Cyclic Redundancy Check

(CRC)http://www.youtube.com/watch?v=Mfby7k6_Z9k&list=PLzVDJSginQzJAujbMbrOfdhoHbihz3yrl&index=1

5.1 Ethernet Protocol

5.1.2 Ethernet MAC Address

MAC Addresses and Hexadecimal

- An Ethernet MAC address is a **48-bit binary value expressed as 12 hexadecimal digits** (4 bits per hexadecimal digit).
- Hexadecimal is used to represent Ethernet MAC addresses and IP Version 6 addresses.
 - Hexadecimal is a base sixteen system using the **numbers 0 to 9 and the letters A to F**.
 - It is easier to express a value as a single hexadecimal digit than as four binary bits.
 - Hexadecimal is usually represented in text by **the value preceded by 0x** (E.g., 0x73).

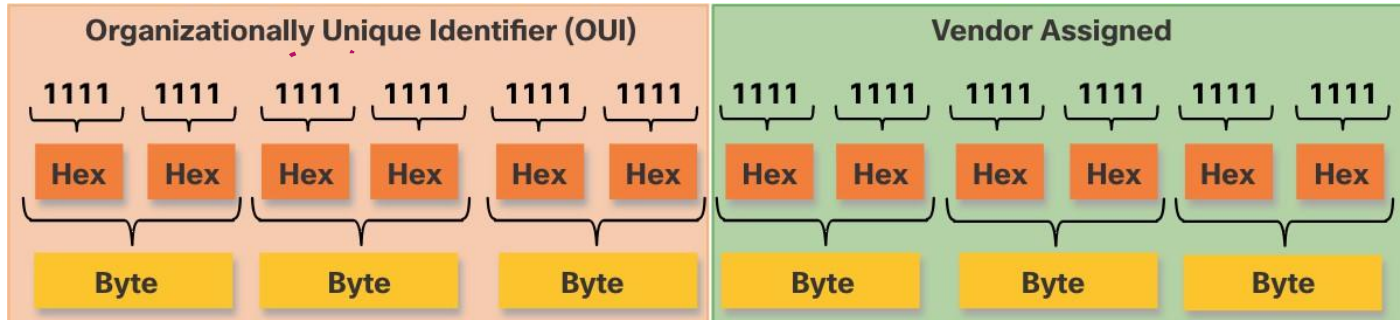
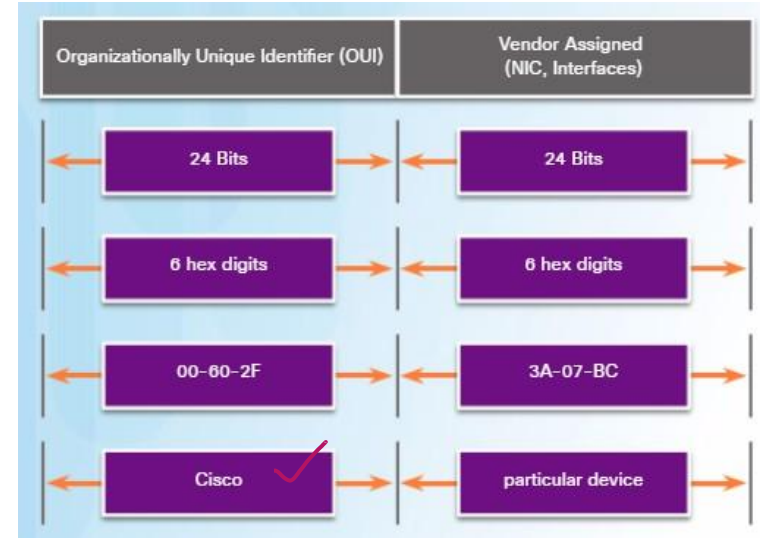
Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Ethernet MAC Address

- In an Ethernet LAN, every network device is connected to the same, shared media. MAC addressing provides a method for device identification at the data link layer of the OSI model.
- An Ethernet MAC address is a 48-bit address expressed using 12 hexadecimal digits. Because a byte equals 8 bits, we can also say that a MAC address is 6 bytes in length.
- All MAC addresses must be unique to the Ethernet device or Ethernet interface. To ensure this, all vendors that sell Ethernet devices must register with the IEEE to obtain a unique 6 hexadecimal (i.e., 24-bit or 3-byte) code called the **organizationally unique identifier (OUI)**.
- An Ethernet MAC address consists of a 6 hexadecimal vendor OUI code followed by a 6 hexadecimal vendor-assigned value.

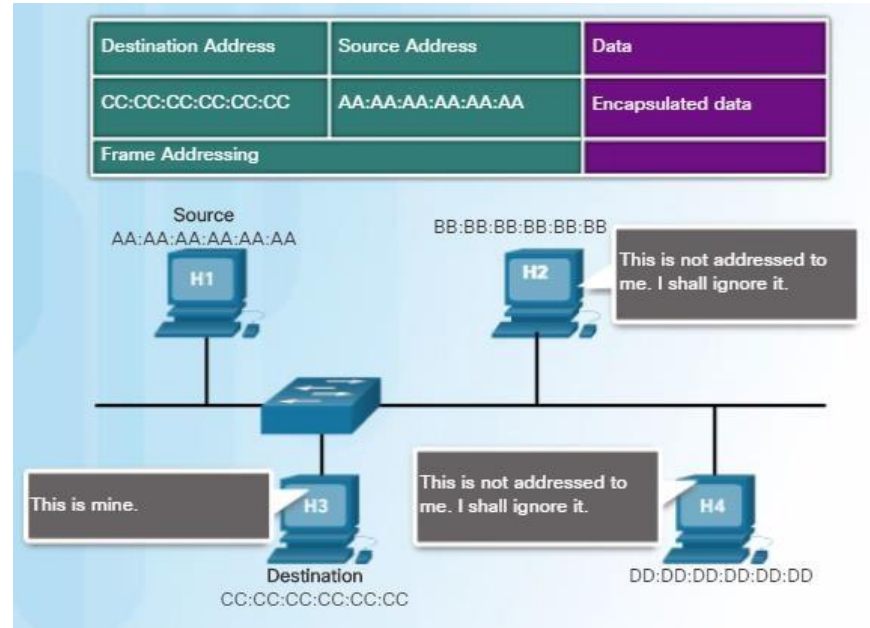
MAC Addresses: Ethernet Identity

- IEEE requires a vendor to follow two simple rules:
 - All MAC addresses assigned to a NIC or other Ethernet device must use that vendor's assigned OUI as the first 3 bytes.
 - All MAC addresses with the same OUI must be assigned a unique value in the last 3 bytes.



Frame Processing

- The MAC address is often referred to as a **burned-in address (BIA)** meaning the address is encoded into the **ROM chip permanently**. When the computer starts up, the first thing the NIC does is copy the MAC address from ROM into RAM.
- When a device is forwarding a message to an Ethernet network, it attaches header information to the frame.
- The header information contains the source and destination MAC address.



MAC Address Representations

- Use the **ipconfig /all** command on a Windows host to identify the MAC address of an Ethernet adapter. On a MAC or Linux host, the **ifconfig** command is used.
- Depending on the device and the operating system, you will see various representations of MAC addresses.

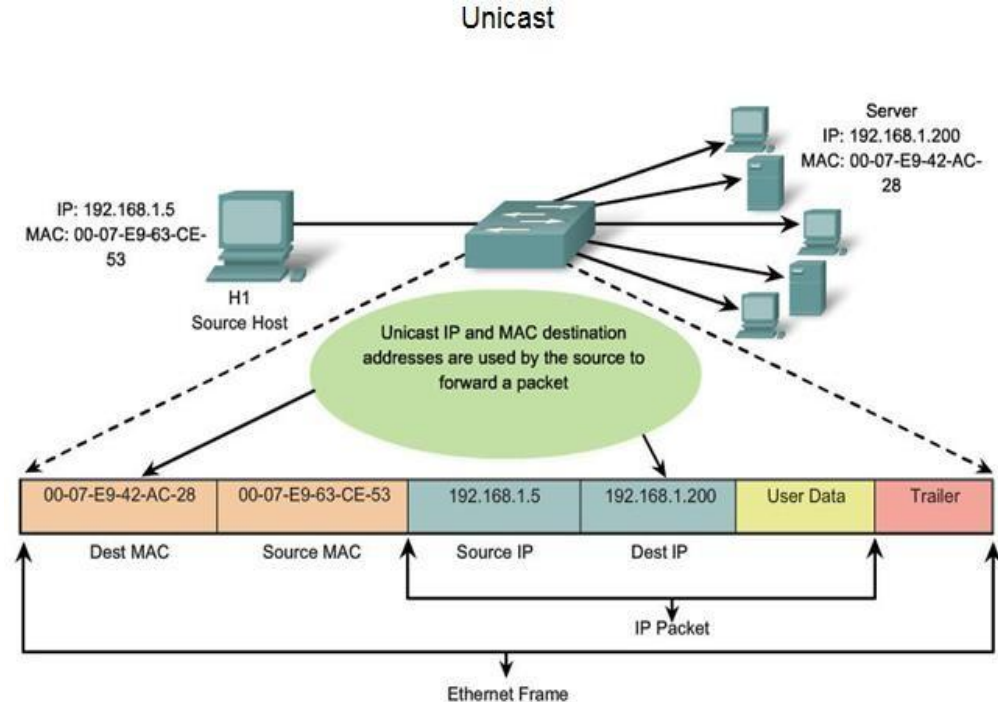


Frame Processing

- MAC addresses assigned to workstations, servers, printers, switches, and routers
- Example MACs: 00-05-9A-3C-78-00, 00:05:9A:3C:78:00, or 0005.9A3C.7800.
- Forwarded message to an Ethernet network, attaches header information to the packet, contains the source and destination MAC address
- Each NIC views information to see if the destination MAC address in the frame matches the device's physical MAC address stored in RAM
- No match, the device discards the frame
- Matches the destination MAC of the frame, the NIC passes the frame up the OSI layers, where the decapsulation process takes place

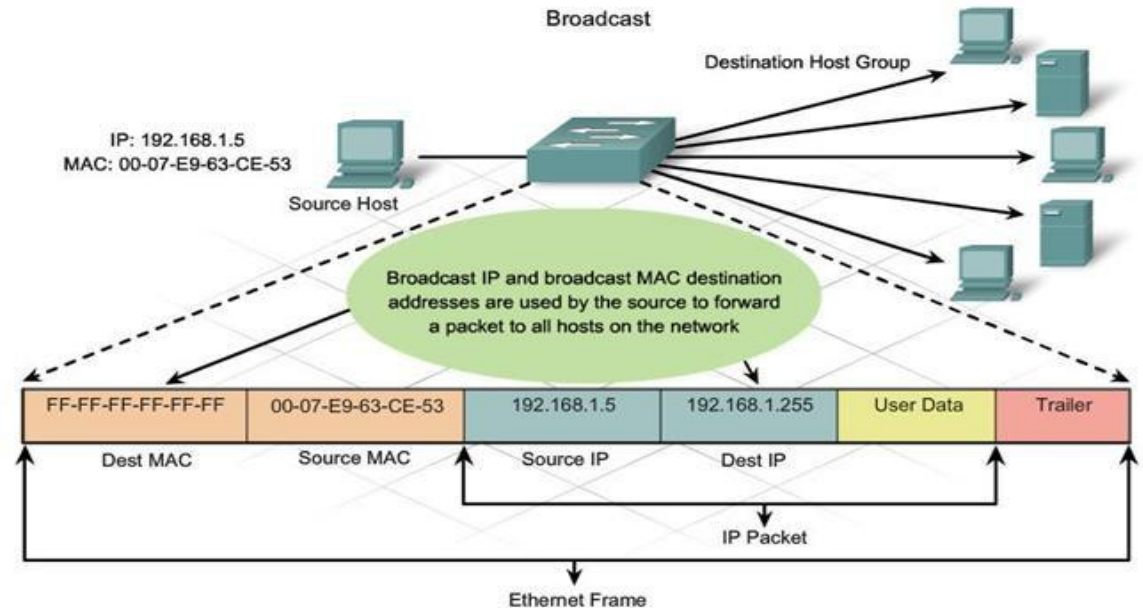
Unicast MAC Address

- A unicast MAC address is the unique address used when a frame is sent from **a single transmitting device to a single destination device**.
- For a unicast packet to be sent and received, a destination IP address must be in the IP packet header and a corresponding destination MAC address must also be present in the Ethernet frame header.



Broadcast MAC Address

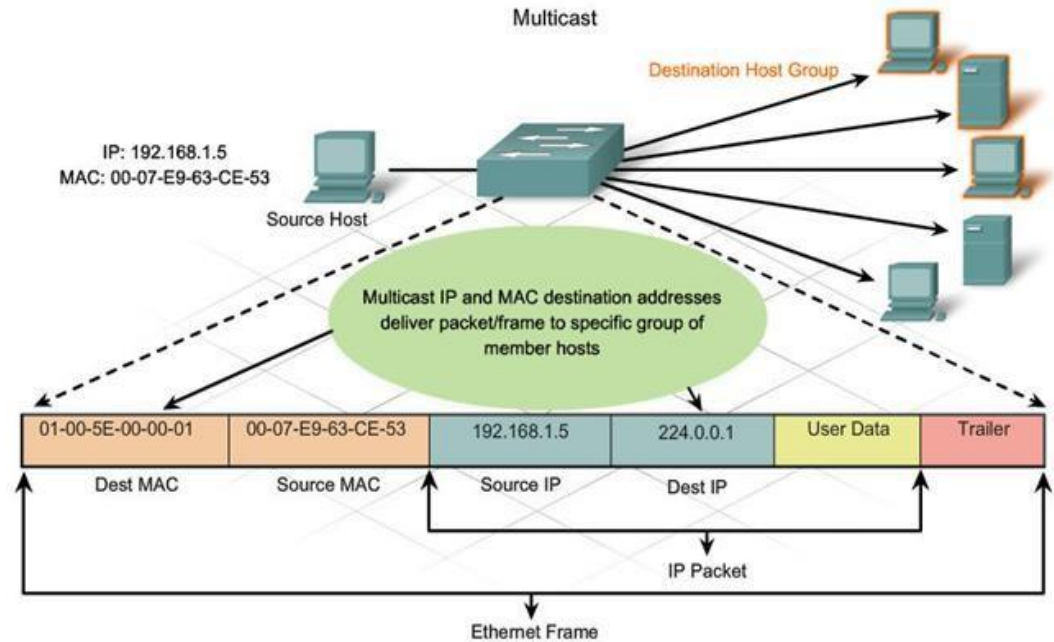
- Many network protocols, such as DHCP and ARP, use broadcasts.
- A broadcast packet contains a destination IPv4 address that has all ones (1s) in the host portion indicating that **all hosts on that local network will receive and process the packet.**



- When the IPv4 broadcast packet is encapsulated in the Ethernet frame, the destination MAC address is the **broadcast MAC address of FF-FF-FF-FF-FF-FF in hexadecimal (48 ones in binary).**

Multicast MAC Address

- Multicast addresses allow a **source device to send a packet to a group of devices.**
- Devices in a multicast group are assigned a multicast group IP address in the range of **224.0.0.0 to 239.255.255.255** (IPv6 multicast addresses begin with FF00::/8).



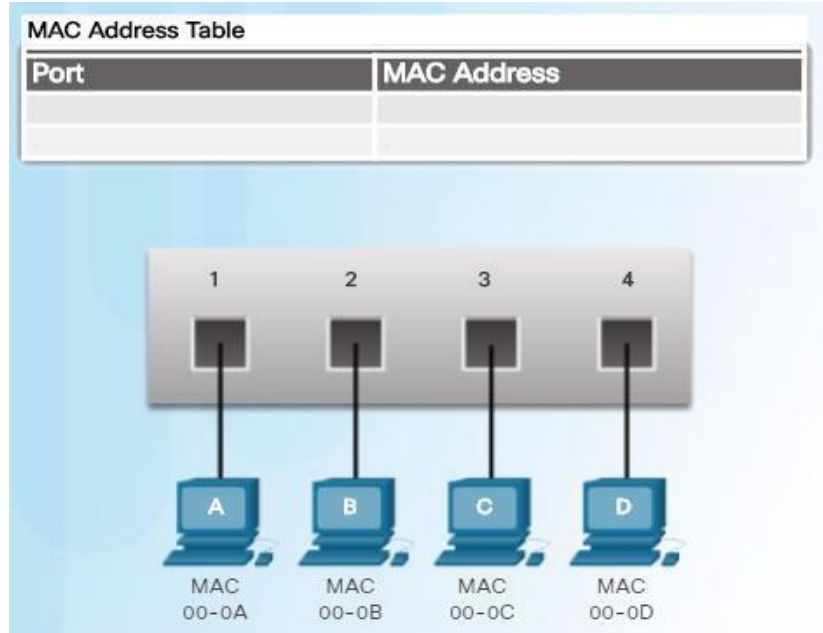
- The multicast IP address requires a corresponding **multicast MAC address that begins with 01-00-5E in hexadecimal.**

5.2 LAN Switches

5.2.1 The MAC Address Table

Switch Fundamentals

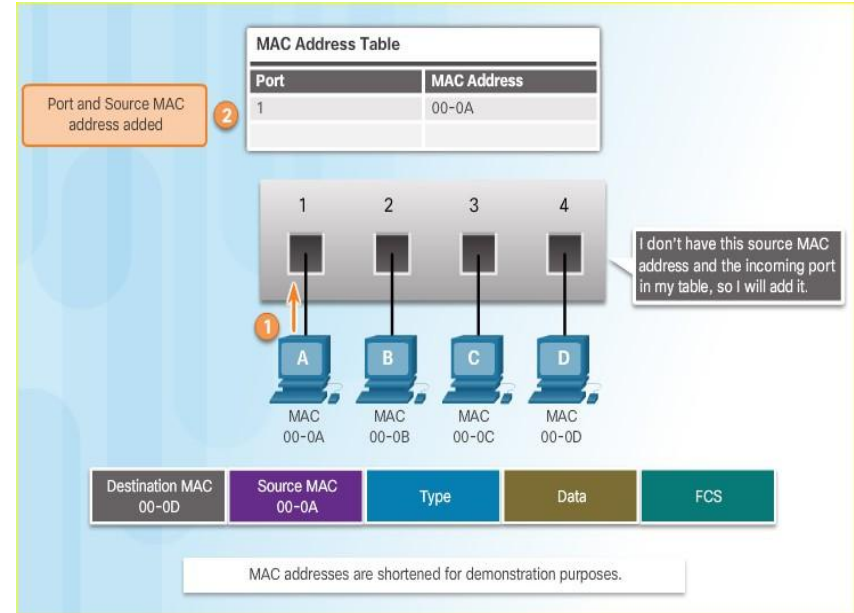
- A Layer 2 Ethernet switch makes its forwarding decisions based only on the Layer 2 Ethernet MAC addresses.
- A switch that is powered on, will have an empty MAC address table as it has not yet learned the MAC addresses for the four attached PCs.
- Note: The **MAC address table** is sometimes referred to as a **content addressable memory (CAM) table**.



Learning MAC Addresses

- The switch dynamically builds the MAC address table. The process to **learn the Source MAC Address** is:

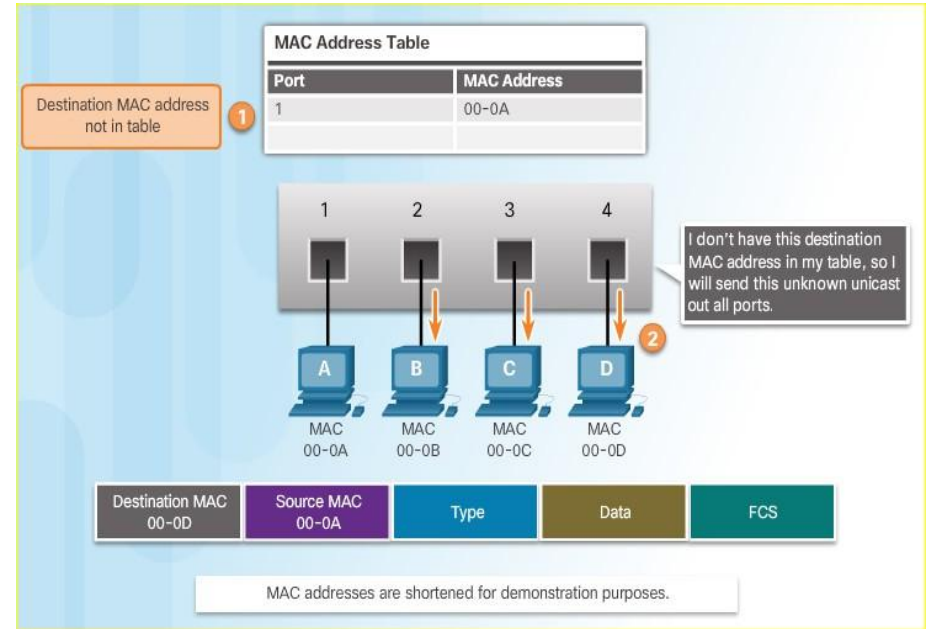
- Switches examine all incoming frames for new source MAC address information to learn.
- If the source MAC address is unknown, it is added to the table along with the port number.
- If the source MAC address does exist, the switch updates the refresh timer for that entry.
- By default, most Ethernet switches keep an entry in the table for 5 minutes.



Learning MAC Addresses (Cont.)

- The process to **forward** the **Destination MAC Address** is:

- If the destination MAC address is a **broadcast or a multicast**, the frame is also **flooded out all ports except the incoming port**.
- If the destination MAC address is a **unicast address**, the switch will look for a **match** in its MAC address table.



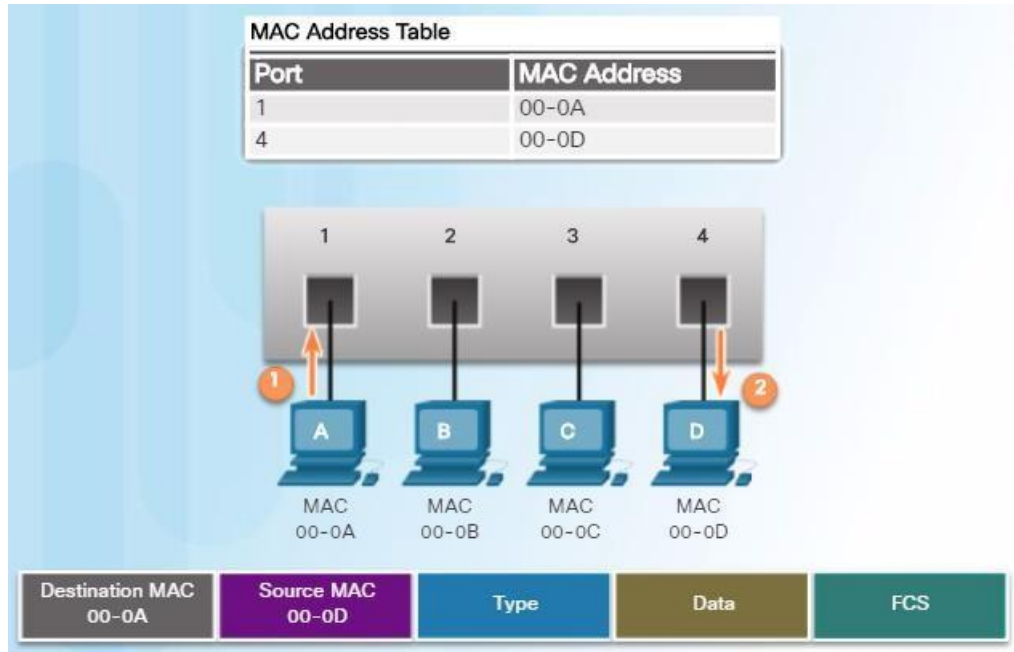
- If the destination MAC address is in the table, it will forward the frame out the specified port.
- If the destination MAC address is not in the table (i.e., an unknown unicast) the switch will forward the frame out all ports except the incoming port.

Learning MAC Addresses

Switching Process	Description
Learn Examining the Source MAC Address	• Switches examine all incoming frames for new source MAC address information to learn. • If the source MAC address is unknown, it is added to the table along with the port number. • If the source MAC address does exist, the switch updates the refresh timer for that entry. • By default, most Ethernet switches keep an entry in the table for 5 minutes.
Forward Examining the Destination MAC Address	• If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port. • If the destination MAC address is a unicast address, the switch will look for a match in its MAC address table. • If the destination MAC address is in the table, it will forward the frame out the specified port. • If the destination MAC address is not in the table (i.e., an unknown unicast) the switch will forward the frame out all ports except the incoming port.

Filtering Frames

- As a switch receives frames from different devices, it is able to populate its MAC address table by examining the source MAC address of every frame.
- When the switch's MAC address table contains the destination MAC address, it is able to filter the frame and forward out a single port.



The MAC Address Table

Video Demonstration - MAC Address Tables on Connected Switches



The MAC Address Table

Video Demonstration - Sending a Frame to the Default Gateway



5.2 LAN Switches

5.2.2 Switch Speeds and Forwarding Methods

Frame Forwarding Methods on Cisco Switches

- Switches use one of the following forwarding methods for switching data between network ports:



Store-and-forward

A store-and-forward switch receives the entire frame, and computes the CRC. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. The frame is then forwarded out the correct port.

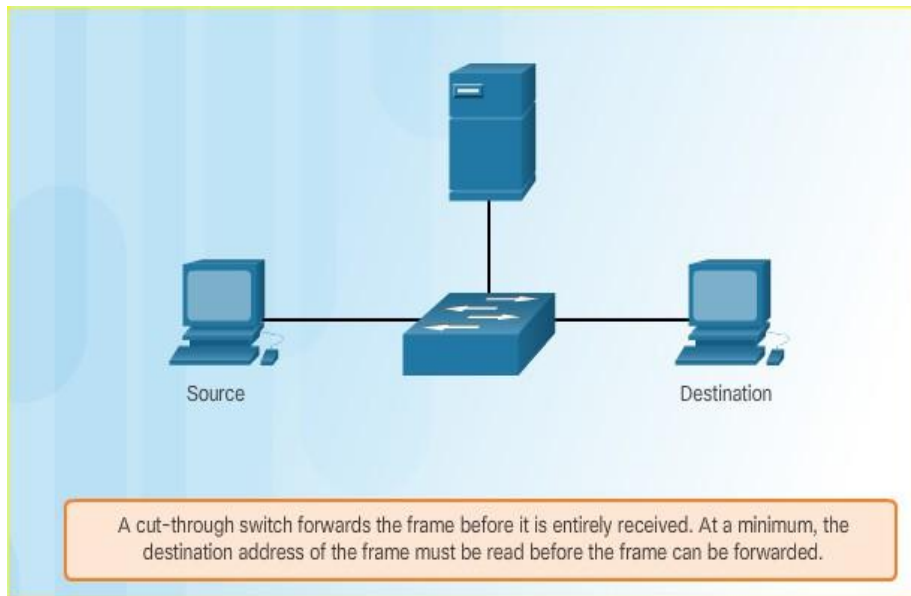


Cut-through

A cut-through switch forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

Cut-Through Switching

- In cut-through switching, the **switch buffers just enough of the frame to read the destination MAC address** so that it can determine to which port to forward the data. The switch **does not perform any error checking on the frame**.
- There are two variants of cut-through switching:



- **Fast-forward switching** offers the lowest level of latency. The switch immediately forwards a packet after reading the destination address. This is the most typical form of cut-through switching.
- **Fragment-free switching**, in which the **switch stores the first 64 bytes of the frame before forwarding**. It is a compromise between store-and-forward and fast-forward switching.

Memory Buffering on Switches

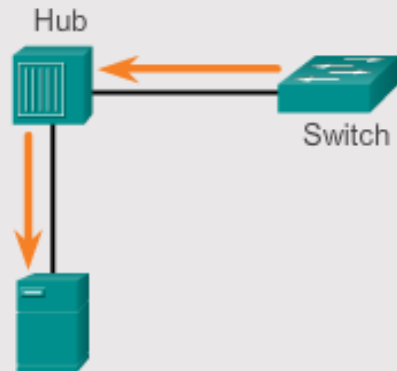
- An Ethernet switch may use a **memory buffering technique** to store frames before forwarding them. Buffering may also be used when the **destination port is busy due to congestion** and the switch stores the frame until it can be transmitted.
- There are two types of memory buffering techniques:

Memory Buffering Method	Description
Port-based memory	• Frames are stored in queues that are linked to specific incoming and outgoing ports. • A frame is transmitted when all the frames ahead of it have been transmitted.
Shared memory	• All frames are deposited into a common buffer which is shared by all ports on the switch.

Duplex Settings

Half Duplex (CSMA/CD)

- Unidirectional data flow
- Higher potential for collision
- Hub connectivity



Full Duplex

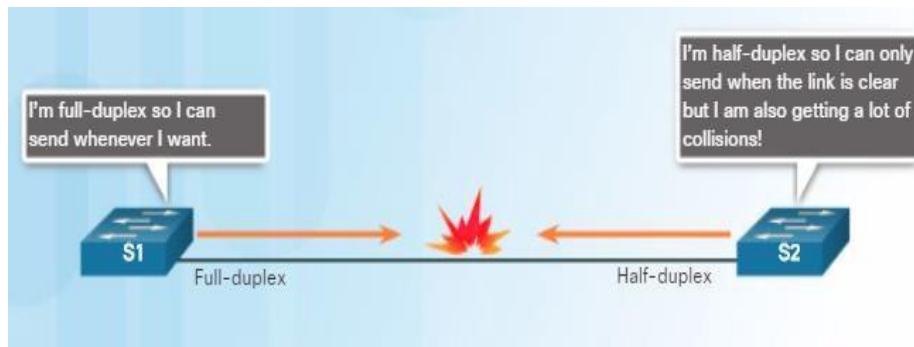
- Point-to-point only
- Attached to dedicated switched port
- Requires full-duplex support on both ends
- Collision-free
- Collision detect circuit disabled



Duplex and Speed Settings

- There are two types of duplex settings used for communications on an Ethernet network:
 - Full-duplex** – Both ends of the connection can send and receive simultaneously.
 - Half-duplex** – Only one end of the connection can send at a time.
- Most devices use autonegotiation which enables two devices to automatically exchange information about speed and duplex capabilities and choose the highest performance mode.

- **Duplex mismatch** is a common cause of performance issues with Ethernet links. It occurs when one port on the link operates at half-duplex while the other port operates at full-duplex.

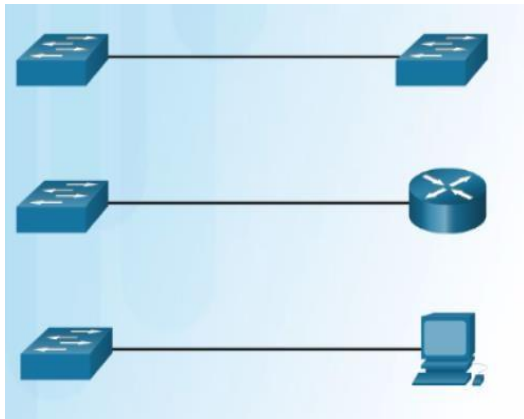


Note: Gigabit Ethernet ports only operate in full-duplex.

Auto-MDIX

- Connections between specific devices such as switch-to-switch, switch-to-router, switch-to-host, and router-to-host devices, once required the use of specific cable types (crossover or straight-through).
- Most switch devices now support the automatic medium-dependent interface crossover (auto-MDIX) feature. This is enabled by default on switches since IOS 12.2(18)SE.
- When enabled using the **mdix auto** interface configuration command, the switch detects the type of cable attached to the port, and configures the interfaces accordingly.

Note: A direct connection between a router and a host requires a cross-over connection.



5.3 Address Resolution Protocol

5.3.1 MAC and IP

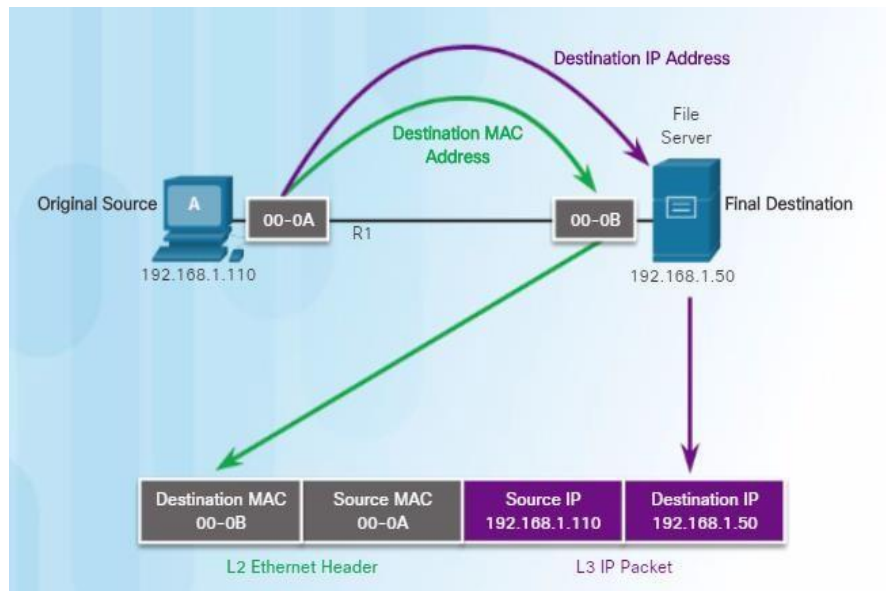
Module 9: Address Resolution

Introduction to Networks v7.02 (ITN)

Destination on Same Network

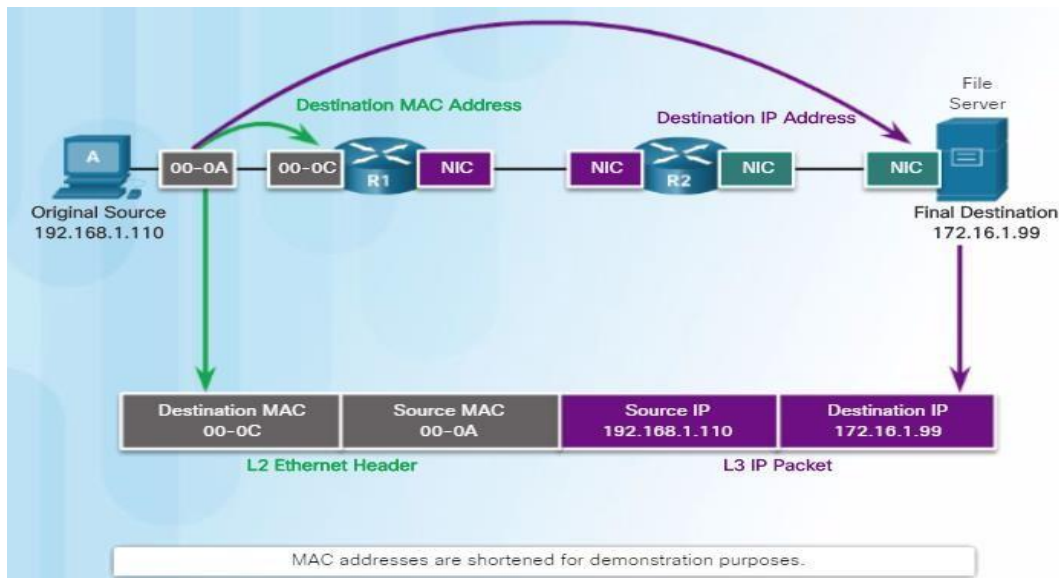
- There are two primary addresses assigned to a device on an Ethernet LAN:
 - **Layer 2 physical address (the MAC address)** – Used for NIC to NIC communications on the same Ethernet network.
 - **Layer 3 logical address (the IP address)** – Used to send the packet from the source device to the destination device.

Layer 2 addresses are used to deliver frames from one NIC to another NIC on the same network. If a destination IP address is on the same network, the destination MAC address will be that of the destination device.



Destination on Remote Network

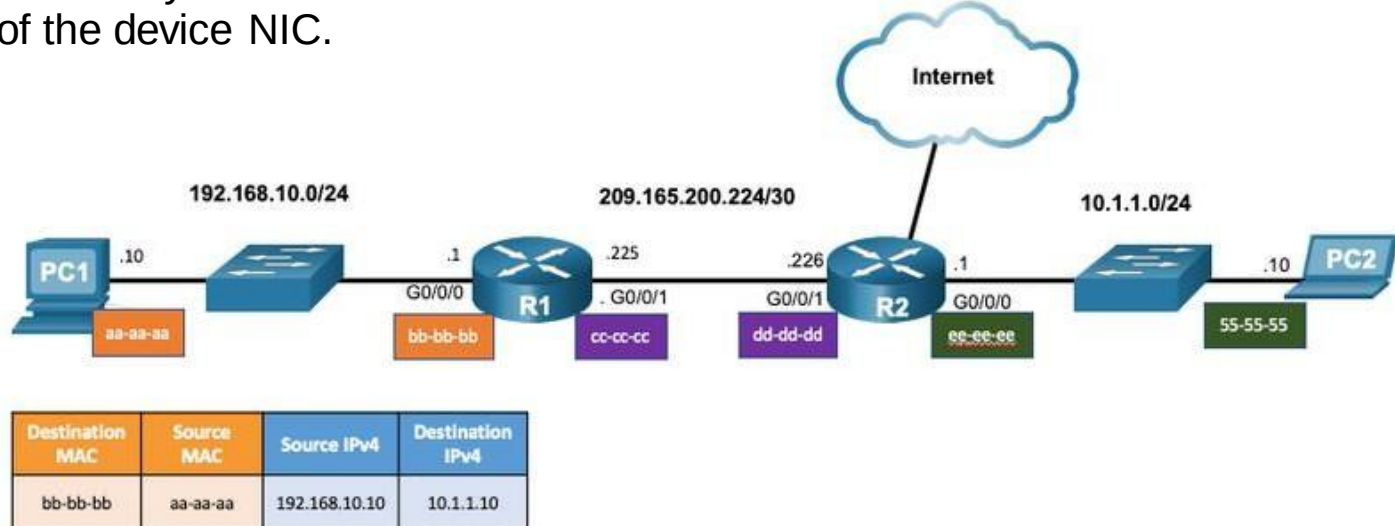
- When the destination IP address is on a **remote network**, the **destination MAC address will be the address of the host's default gateway**.
- In the figure, PC-A is sending an IP packet to a web server on a remote network. The destination IP address is that of the File Server. The destination MAC address is that of Ethernet interface of R1.



Destination on Remote Network

When the destination IP address is on a remote network, the destination MAC address is that of the default gateway.

- **ARP** is used by IPv4 to associate the IPv4 address of a device with the MAC address of the device NIC.
- **ICMPv6** is used by IPv6 to associate the IPv6 address of a device with the MAC address of the device NIC.



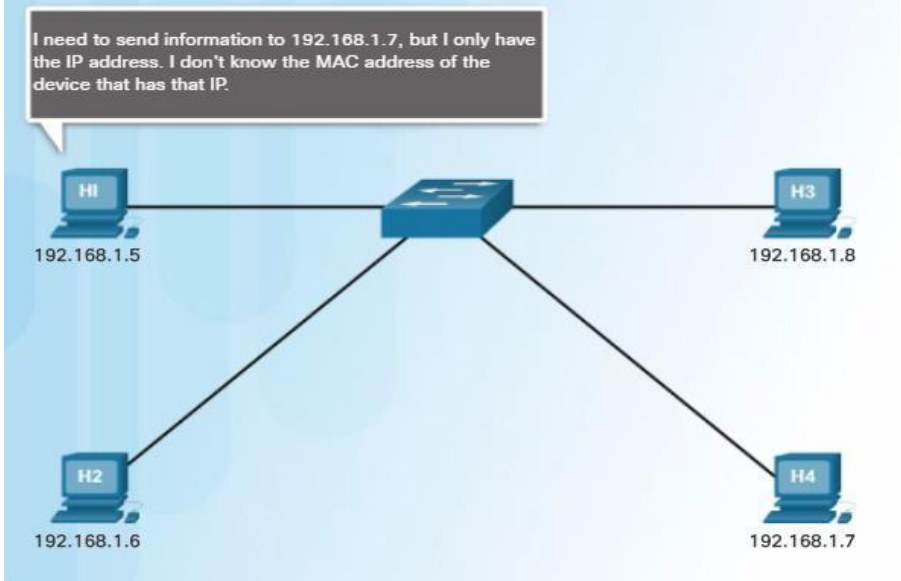
5.3 Address Resolution Protocol

5.3.2 ARP

Module 9: Address Resolution
Introduction to Networks v7.0 (ITN)

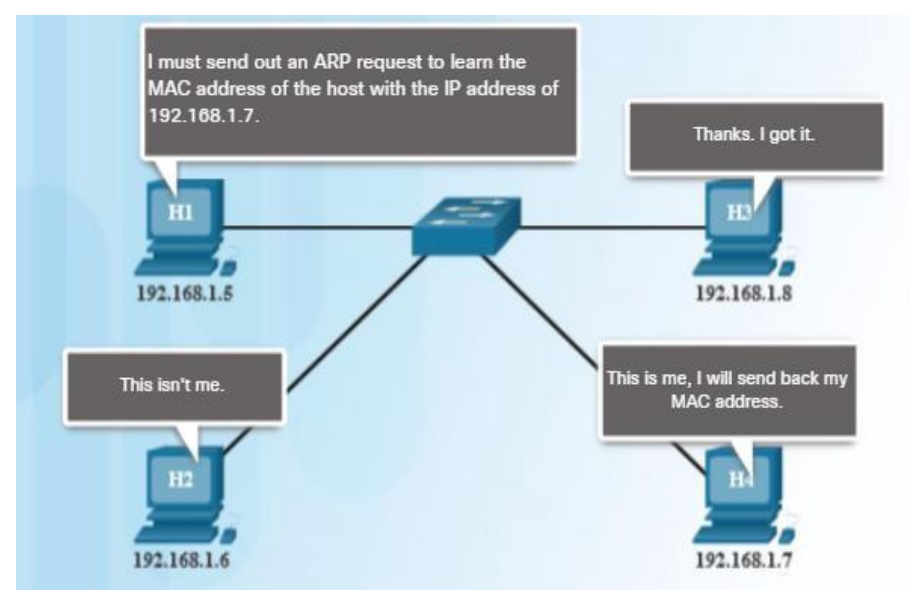
Introduction to ARP

- When a device sends an Ethernet frame, it contains these two addresses:
 - Destination MAC address
 - Source MAC address
- ARP Purpose
 - Sending node needs a way to find the MAC address of the destination for a given Ethernet link
 - To determine the destination MAC address, the device uses ARP.
- ARP provides **two basic functions**:
 - **Resolving IPv4 addresses to MAC addresses**
 - **Maintaining an ARP table of IPv4 to MAC address mappings**



ARP Functions

- Ethernet devices refer to an **ARP table** (or the **ARP cache**) in its memory (i.e., RAM) to find the MAC address that is mapped to the IPv4 address.
- A device will search its ARP table for a destination IPv4 address and a corresponding MAC address.
 - If the packet's destination IPv4 address is on the same network as the source IPv4 address, the device will search the ARP table for the destination IPv4 address.
 - If the destination IPv4 address is on a different network than the source IPv4 address, the device will search the ARP table for the IPv4 address of the default gateway.

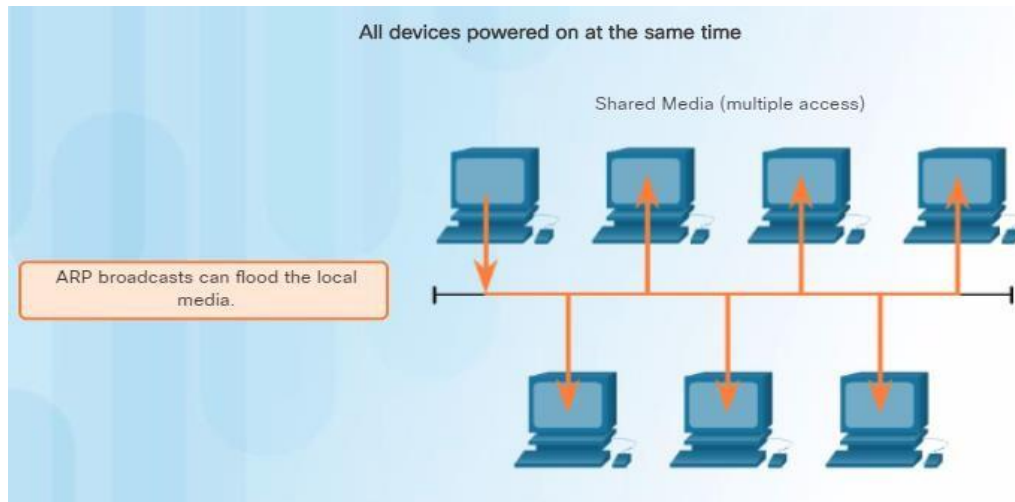


Video Demonstration – ARP Request

- An **ARP request is a broadcast frame** sent when a device needs a MAC address associated with an IPv4 address, and it does not have an entry for the IPv4 address in its ARP table.
- ARP messages are encapsulated directly within an Ethernet frame. There is no IPv4 header.
- The ARP request message includes:
 - Target IPv4 address
 - Target MAC address

ARP Broadcasts

- As a broadcast frame, an ARP request is received and processed by every device on the local network.
- ARP requests can flood the local segment if a large number of devices were to be powered up and all start accessing network services at the same time.



ARP Request – Video



Video Demonstration – ARP Reply

- Only the device with an IPv4 address associated with the target IPv4 address in the ARP request will respond with an ARP reply.
- The ARP reply message includes:
 - Sender's IPv4 address
 - Sender's MAC address
- Entries in the ARP table are time stamped. If a device does not receive a frame from a particular device by the time the timestamp expires, the entry for this device is removed from the ARP table.

ARP Reply – Video



ARP Role in Remote Communication

- If the destination IPv4 host is on the local network, the frame will use the MAC address of this device as the destination MAC address
- If the destination IPv4 host is not on the local network, the source uses the ARP process to determine a MAC address for the router interface serving as the gateway
- In the event that the gateway entry is not in the table, an ARP request is used to retrieve the MAC address associated with the IP address of the router interface

Video Demonstration – ARP role in Remote Communications

- When a host creates a packet for a destination, it compares the destination IPv4 address and its own IPv4 address to determine if the two IPv4 addresses are located on the same Layer 3 network.
- If the destination IPv4 host is on the local network, the frame will use the MAC address of this device as the destination MAC address
- If the destination IPv4 host is not on the local network, the source uses the ARP process to determine a MAC address for the router interface serving as the gateway
- In the event that the gateway entry is not in the table, an ARP request is used to retrieve the MAC address associated with the IP address of the router interface

ARP

ARP Role in Remote Communication – Video



Demonstration

ARP Role in Remote Communication

Removing Entries from an ARP Table

- Entries in the ARP table are not permanent and are removed when an ARP cache timer expires after a specified period of time.
- The duration of the ARP cache timer differs depending on the operating system.
- ARP table entries can also be removed manually by the administrator.

ARP Tables on Networking Devices

- The **show ip arp** command displays the ARP table on a Cisco router.
- The **arp -a** command displays the ARP table on a Windows 10 PC.

```
R1# show ip arp
```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.10.1	-	a0e0.af0d.e140	ARPA	GigabitEthernet0/0/0

```
C:\Users\PC> arp -a
```

```
Interface: 192.168.1.124 --- 0x10
```

Internet Address	Physical Address	Type
192.168.1.1	c8-d7-19-cc-a0-86	dynamic
192.168.1.101	08-3e-0c-f5-f7-77	dynamic

ARP Issues – ARP Broadcasting and ARP Spoofing

- ARP requests are received and processed by every device on the local network.
- Excessive **ARP broadcasts** can cause some reduction in performance.
- For example, if a large number of devices were to be powered up and all start accessing network services at the same time.
- ARP can lead to a potential security risk. A threat actor can use **ARP spoofing** to perform an ARP poisoning attack.

ARP Spoofing

- Attackers can respond to requests and pretend to be providers of services.
- One type of ARP spoofing attack used by attackers is to reply to an ARP request for the default gateway. In the figure, host A requests the MAC address of the default gateway. Host C replies to the ARP request. Host A receives the reply and updates its ARP table. It now sends packets destined to the default gateway to the attacker host C.
- Enterprise level switches include mitigation techniques known as **dynamic ARP inspection (DAI)**.

